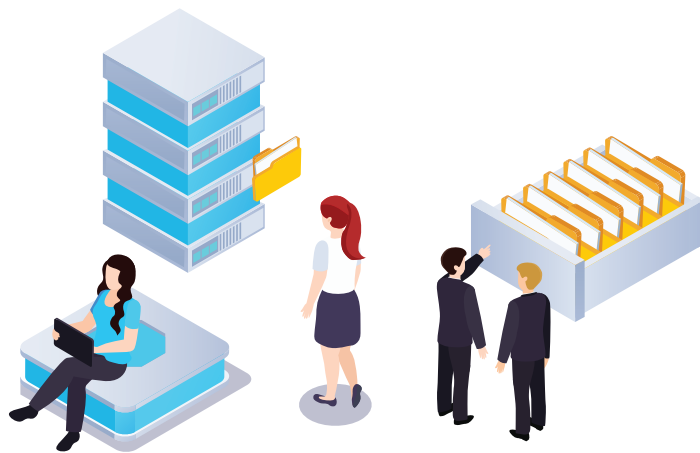


Apache Druid: A Powerful Real-time Analytics Database



We generate humongous amounts of data these days, and therefore need specific techniques to store and handle it. Apache Druid is a powerful and modern analytics database that is really fast and allows users to conduct real-time data analyses. Designed specifically for processing real-time event-driven data, it is a strong platform that combines the capability of time-series databases, column-oriented analytics databases and search systems.



In today's world, data is the lifeblood of any software product. Every time we engage with a digital system, we generate data. The amount of data available has grown to the point where specialist tools are required to make sense of it. Real-time analytics allows for the extraction of insights from large amounts of data. To undertake this type of analysis, we need custom-built tools that can store and process data in real time, allowing us to make real-time inferences.

Apache Druid is not simply a traditional data warehouse but is tailor-made to handle real-time analytics of event-driven data. Druid is used to power the graphical user interface (GUI) of analytics applications.

It serves as the backend for providing the data manipulation for powering highly concurrent queries. Druid can be thought of as a combination of three major things (Figure 1):

- Time series databases
- Column oriented databases
- Search systems

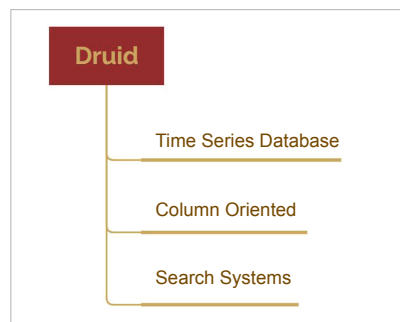


Figure 1: Apache Druid

Druid has applications across various domains. Almost all domains generate massive scales of data and Druid can be effectively deployed to analyse these. Some of the applications of Druid are (Figure 2):

- Click stream, view stream and activity stream analytics can be carried out with Druid. User activity and behavioural analytics can be performed too
- Performance monitoring of networks with the help of network telemetry analytics
- Server metrics storage
- Supply chain analytics
- Analytics of digital marketing and advertising. Druid can be effectively implemented to query online advertising data. This can be used to